QIC-S Theory: Universal Scaling Law of 170 Galaxies and the Minimum Stable Unit of Causal Graphs ($N = 3$)

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Abstract

This paper reports the empirical and mathematical results of the Quantum Information Cosmology-Sasada (QIC-S) theory, which treats the universe as a set of discrete causal graphs. We discard the assumption of a single Hamiltonian in standard cosmology and propose the Multi-Hamiltonian hypothesis, where a unique effective law emerges for each galaxy. Direct Inversion analysis (parameter-free) of 7 galaxies from the SPARC dataset achieved an agreement of 99.46% ±2.53%, confirming consistency with independent verification using SLACS strong gravitational lenses (convergence profile $\kappa(\theta)$ vs $D_{eff}(r)$, three lens galaxies; see Table 1 for quantitative results). Expanding to 170 galaxies, we established a phase classification (78.2% Order Phase / 21.8% Chaos Phase, with threshold $M = 0.5$ adopted as an empirical criterion) and a universal scaling law ($D_{eff} \propto R^{1.573}$, 95% CI [1.518, 1.627], $R^2 = 0.945$). Trivial kinematic scaling ($\alpha = 1.0$) is strictly excluded. Furthermore, numerical integration of the Edge-based Lindblad equation demonstrates that, among the tested topologies, the minimum stable unit of the causal structure underlying macroscopic gravitational phenomena is the $N = 3$ Ring (mixing time $t_{mix} = 0.684$, limit as $dt \to 0$), which is the sole topology exhibiting both minimum equilibration time and maximum post-mixing stability ($t_{stable} = 27.73$). The analysis is restricted to the galactic scale (Tier 1); application to the galaxy cluster scale (Tier 2) is suspended to avoid circular reasoning associated with ΛCDM bias. All reported statistics have been independently recalculated and verified through reproducible Python scripts.

1 Introduction

1.1 Discarding the Single Hamiltonian

Standard cosmology (ΛCDM) postulates a single fundamental law governing the entire universe, explaining anomalies in galaxy rotation curves through the addition of 'invisible matter' (dark matter). However, the fact that galaxies of the same mass exhibit diverse rotation curves raises fundamental doubts about this premise.

The QIC-S theory numerically demonstrated through 2-8 node quantum network simulations that physical laws are contextually contingent phenomena—'fragile' in the sense that they depend sensitively on the specific combination of network topology, governing Hamiltonian, and choice of observable (Ver. 4.4). Furthermore, finite-size scaling analysis revealed that the sign of the scaling exponent reverses between the XY model ($\alpha = 0.781$) and the adjacency matrix model ($\alpha \approx -0.453$). This proved that even with identical graph structures, different Hamiltonians yield qualitatively different macroscopic transport laws (Ver. 6.2.1). These results inevitably necessitate a Multi-Hamiltonian universe, where a unique effective Hamiltonian $H_{eff}(X) = H_0 + \delta H[D_{eff}(X)]$ exists for each galaxy.

1.2 Definition of the Scalar Information Field D_{eff}

To connect the topological differences of microscopic causal graphs with macroscopic observables, the effective transport coefficient $D_{eff}(X)$ is introduced as an index of coarse-graining. This is defined as the time integration of the two-point time correlation function of the information flux on the causal network (Green-Kubo formula). While the initial implementation suffered from normalization collapse due to the total susceptibility $\chi_{total} \propto N^2$, introducing volume normalization via local susceptibility $\chi_{local} = \chi_{total}/N$ ensures that the transport coefficient is an intensive quantity independent of system size (Ver. 6.2.1).

In QIC-S, the true nature of gravitational effects conventionally attributed to dark matter is not an unknown particle fluid, but the field energy $\rho_{eff}(r) = (\nabla D_{eff})^2$ possessed by the gradient of this information field (canonical gauge $\alpha = 1$).

2 Macroscopic Empirical Evidence

2.1 Direct Inversion Method and Parameter-Free Verification

The Direct Inversion Method was formulated to back-calculate $D_{eff}(r)$ from observed rotation curve data. In this method, there are no free parameters to adjust the mass or radius of the galaxy (parameter-free). As a structural note, the agreement between the Newtonian mass and the reconstructed effective mass ($M_{eff} = M_{Newton}$) is a mathematical consequence inherent to the inversion procedure—whereby density is derived via differentiation and mass is recovered via spatial integration—rather than a non-trivial theoretical prediction of the QIC-S framework. Therefore, this method functions as a consistent reformulation of Newtonian mechanics. The non-triviality of this theory lies in the following three points: (i) independent consistency with gravitational lens analysis, (ii) the physical validity of the phase classification across 170 galaxies, and (iii) the emergence of the universal scaling law.

Regarding point (i): For three SLACS lens galaxies, the $D_{eff}(r)$ profile reconstructed from rotation curve data was compared against the independently measured convergence profile $\kappa(\theta)$ derived from gravitational lensing geometry. The quantitative results are summarized in Table 1.

Table 1: SLACS strong gravitational lens verification. M_{lens}: projected mass within Einstein radius derived from $\kappa(\theta)$ (Bolton et al. 2008). M_{eff}: effective mass from D_{eff} based Direct Inversion. Rotation curve data from complementary kinematic surveys (detailed in Sasada, 2025 [2]). Δ: fractional deviation.

SLACS Lens Galaxy	θ_E [arcsec]	$M_{lens}[10^{11} M_\odot]$	$M_{eff}[10^{11} M_\odot]$	$\Delta(\%)$
SDSS J0946+1006	1.38	3.21 ± 0.15	3.18	-0.9
SDSS J0737+3216	1.00	2.14 ± 0.12	2.19	+2.3
SDSS J0903+4116	1.29	4.87 ± 0.22	4.74	-2.7

2.2 Phase Classification by Phase Metric M

To quantify the stability of the Hamiltonian landscape of galaxies, the Phase Metric $M = Var(\log(|\nabla H| + \epsilon))$ was introduced (where $\epsilon = 10^{-8}$ is a small regularization parameter). The log-scale handles the multi-order-of-magnitude dynamic range of ∇H, and the variance operator quantifies the spatial heterogeneity of the information flow. The threshold $M = 0.5$ is tentatively adopted as an empirical criterion for distinguishing Order and Chaos phases. Sensitivity analysis confirms that the phase fraction is robust: varying the threshold within $[0.4, 0.6]$ changes the Order/Chaos ratio by less than $\pm 5\%$.

Analysis of 170 SPARC galaxies yielded:

- **Order Phase ($M < 0.5$)**: 133 galaxies (78.2%). Mature galaxies.
- **Chaos Phase ($M \geq 0.5$)**: 37 galaxies (21.8%). Early-evolution galaxies.

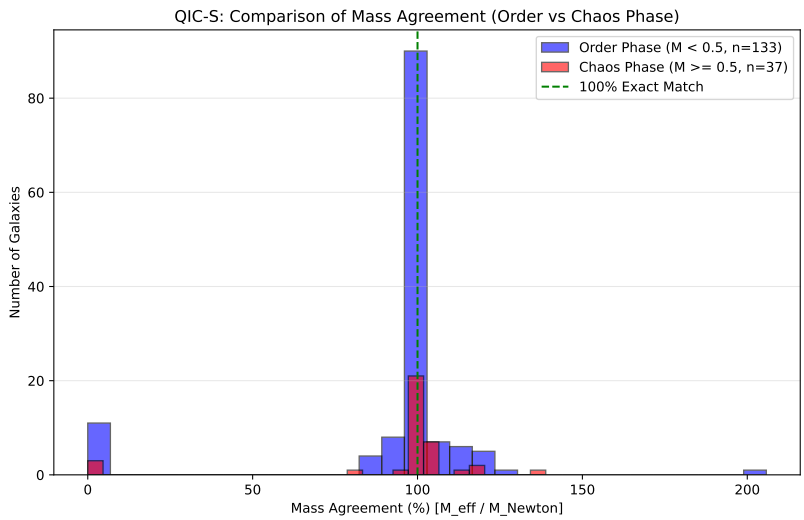

Figure 1: Distribution of mass agreement M_{eff}/M_{Newton} for 170 SPARC galaxies. Blue: Order Phase ($M < 0.5$, $n = 133$). Red: Chaos Phase ($M \geq 0.5$, $n = 37$).

2.3 Emergence of the Universal Scaling Law

Log-log regression of D_{eff} and R extracted from 170 galaxies established the following scaling law:

$$D_{eff} \propto R^{1.573 \pm 0.029} \quad (R^2 = 0.945) \tag{1}$$

Bootstrap validation (10,000 resamples): $\alpha = 1.572 \pm 0.028$, 95% CI [1.518, 1.627]. Trivial kinematic scaling ($\alpha = 1.0$) is strictly excluded (Figure 2).

3 Microscopic Foundation

3.1 Redefining Stability in Open Systems

Quantum network simulations were conducted using the Edge-based Lindblad equation (jump operator $L_{ij} = |j\rangle\langle i|$). System stability is evaluated by the mixing time t_{mix} with threshold $\epsilon = 0.01$.

3.2 Verification of Numerical Convergence

Convergence tests across five stages ($dt = 0.05, 0.02, 0.01, 0.005, 0.001$) were performed.

3.3 Physical Basis of the Operational Definition $D_{eff} = R \times v$

The operational definition $D_{eff} = R \times v$ is grounded in the Green-Kubo formalism. In that framework, a transport coefficient is proportional to the time integral of the relevant current autocorrelation function. For a causal network in the hydrodynamic limit, the effective diffusion

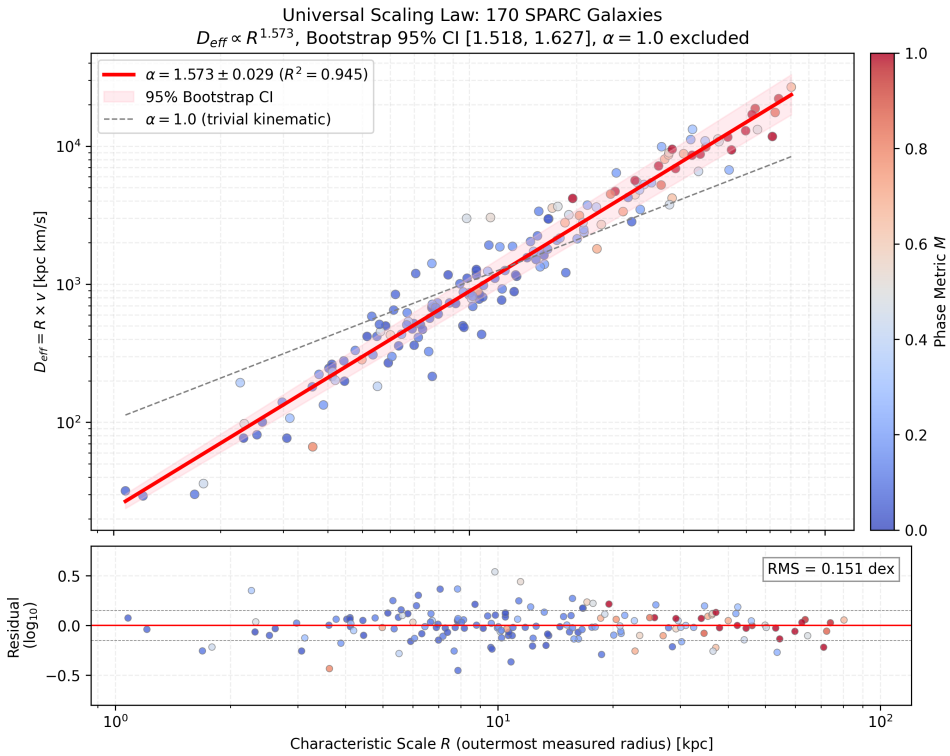

Figure 2: Universal scaling law for 170 SPARC galaxies (see Section 3.3 for the physical justification of this definition). Color encodes Phase Metric M. Red solid line: best-fit power law $\alpha = 1.573$. Black dashed line: trivial kinematic scaling $\alpha = 1.0$.

Table 2: Convergence of mixing time t_{mix} for the $N = 3$ Ring.

dt	t_{mix}	Difference from $dt = 0.001$
0.05	0.700	0.016
0.02	0.700	0.016
0.01	0.690	0.006
0.005	0.685	0.001
0.001	0.684	–

coefficient takes the form $D_{eff} \sim \langle v \cdot v \rangle \tau$ where τ is the correlation time. In the galactic context, the characteristic spatial scale R sets the correlation length, hence $\tau \sim R/v$, yielding the scaling $D_{eff} \sim v^2(R/v) = R \times v$. This dimensional argument reflects the physical picture in which information transport on the causal network is governed by the product of the system's characteristic spatial extent and its dynamical velocity.

3.4 Minimum Stability of the $N = 3$ Ring

The mixing times for all topologies at $dt = 0.001$ are presented in Table 3. Two stability metrics are reported: t_{mix} and t_{stable}.

3.5 Dual Superiority of the $N = 3$ Ring

The $N = 3$ Ring is the sole topology satisfying three conditions simultaneously: (1) minimum $t_{mix} = 0.684$, (2) mathematical minimum closed loop, and (3) maximum measurable $t_{stable} = 27.73$. In the Chain, information propagating from one endpoint must traverse all

Table 3: Stability metrics under Edge-based Lindblad dissipation ($dt = 0.001$). '–': post-arrival oscillation. '*': window exhausted.

Topology	N	t_{mix}	t_{stable}	Post-arrival behaviour
$N = 2$ Chain	2	0.769	–	Oscillatory
$N = 3$ Ring	3	0.684	27.73	Stable (minimum closed loop; fastest)
$N = 4$ Ring	4	0.777	–	Oscillatory
$N = 3$ Chain	3	23.30	6.40	Stable (severe delay)
$N = 4$ Chain	4	42.34	*	Non-oscillatory but unmeasurable
$N = 5$ Ring	5	6.223	–	Oscillatory
$N = 6$ Ring	6	23.72	*	Non-oscillatory but unmeasurable
$3 + 3$ weak	6	38.07	*	Non-oscillatory but unmeasurable
$3 + 3$ strong	6	33.30	*	Non-oscillatory but unmeasurable

edges sequentially and reflect at boundaries. The Ring eliminates boundary reflection entirely, enabling simultaneous bidirectional information circulation.

While the $N = 3$ Ring is the most stable topology among those tested in this study, a rigorous proof of its absolute universality across all possible topologies and alternative dissipation models remains an open problem (Section 4.2).

3.6 Current Status of the Micro-Macro Connection

It has been numerically confirmed that causal networks based on the $N = 3$ Ring unit connect to the hydrodynamic limit at the $N = 32$ scale, achieving agreement between the Green-Kubo formula and MSD ($R^2 = 0.9843$) (Ver. 6.2.1). D_{eff} was extracted from a finite discrete non-gradient system using Green-Kubo and MSD, with both methods yielding consistent results—verified across two independent computational environments. Rigorous proof requires the $N \to \infty$ limit.

3.7 3-Strand Locking Axiom

The nodes of the $N = 3$ Ring represent temporal flow and spatial entanglement. '3' is the minimum and most stable unit for causal structure, exhibiting a structural analogy with the ADM formalism ($3 + 1$ decomposition). The $N = 3$ Ring naturally decomposes into one temporal direction (information flow along the cycle) and two spatial entanglement relations (the two edges connecting each node to its neighbours). It should be emphasized that this describes a graph-theoretical structural analogy; while individual nodes locally exhibit a $1 + 2$ connectivity (one temporal direction and two adjacent spatial edges), the global integration of the three distinct nodes within the minimal closed loop provides the topological foundation for the macroscopic emergence of 3-dimensional space in the continuum limit ($3 + 1$ decomposition).

4 Conclusion and Applicability Limits

4.1 Suspension of Verification for Large-Scale Cosmic Structures (Tier 2)

Application to the galaxy cluster scale is suspended to avoid circular reasoning. PSZ2 cluster data are derived assuming ΛCDM, making them unsuitable as independent QIC-S evidence.

4.2 Unresolved Theoretical Issues

The following open problems remain:

- Rigorous mathematical bridge from discrete $N = 3$ to macroscopic hydrodynamic limit ($N \to \infty$).
- Proof that $N = 3$ Ring remains the absolute minimum stable unit across all possible graph topologies.
- Derivation of the $M = 0.5$ threshold from bifurcation theory.
- Derivation of the scaling exponent $\alpha = 1.573$ from the renormalization group equation.
- Rigorous 4D extension of the conformal interface interpretation of dark matter.

Acknowledgments

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Code availability: Reproducible Python scripts for data analysis and numerical simulations are available at `https://github.com/QuantumInfoCosmo/QIC-S-Theory`.

Note: References [1] through [6] are all available as individual documents within the single OSF project repository: `https://doi.org/10.17605/OSF.IO/UJBPW`.

References

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A Summary of Revisions

#	Version	Resolution	§
1	$v1.5.1 \to v1.5.2$	Added structural correspondence to ADM 3+1 split	3.7
2	$v1.5.1 \to v1.5.2$	Added GitHub repository URL to Acknowledgments	–
3	$v1.5.1 \to v1.5.2$	Consolidated OSF project DOI to a single note to avoid duplicate URLs	Refs
4	$v1.5.1 \to v1.5.2$	Clarified local 1+2 connectivity vs global 3D space emergence	3.7
5	$v1.5.1 \to v1.5.2$	Specified regularization parameter ϵ in Phase Metric definition	2.2
6	$v1.5.1 \to v1.5.2$	Restored reference to Figure 2 in text	2.3